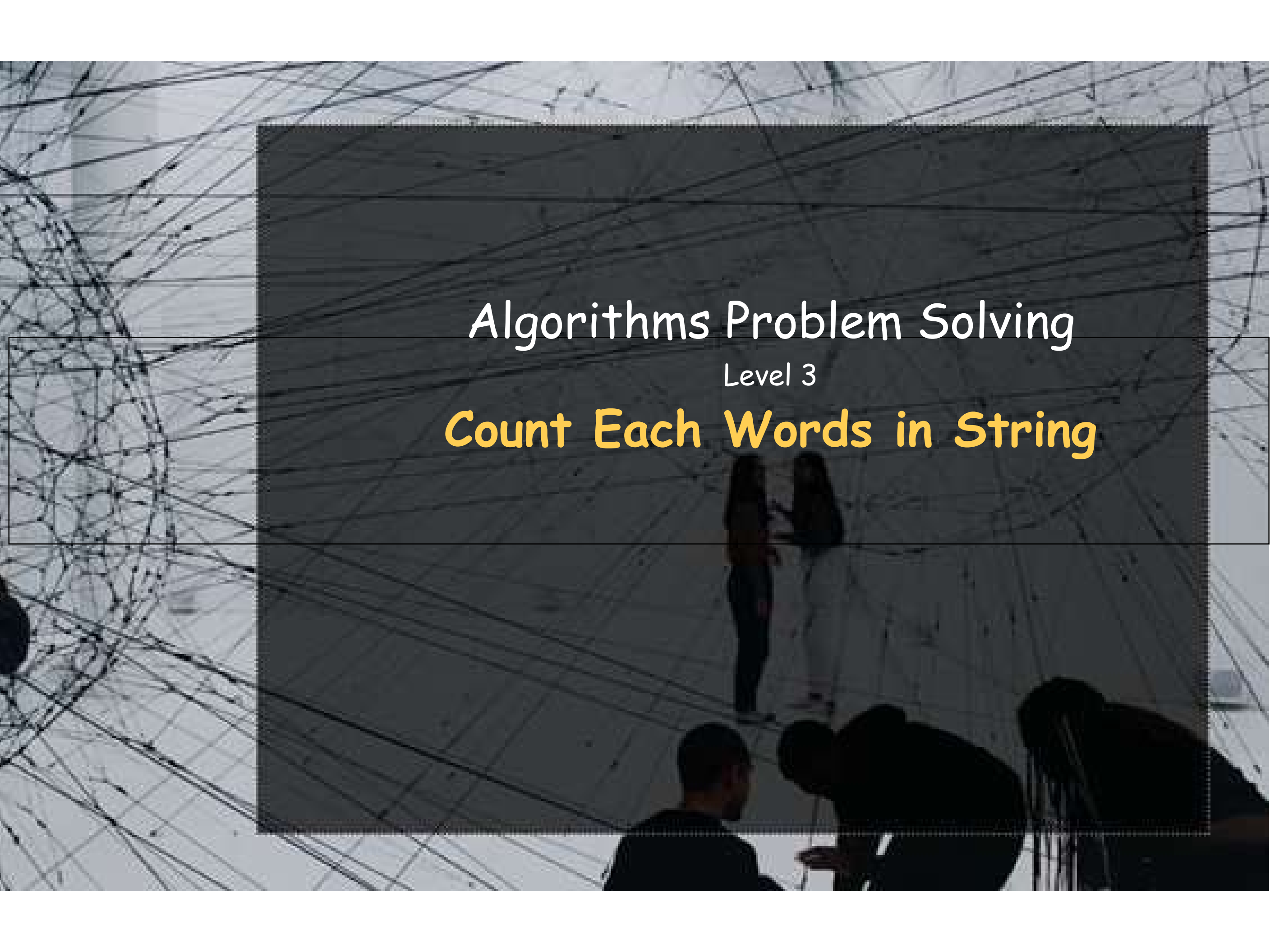


Algorithms Problem Solving

Level 3

Count Each Words in String



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Definition

Word Count Calculation

-  Program determines how many words exist in a sentence.

-  It can reuse `SplitString()` to convert the sentence into tokens and count them.

Explanation

Tokenization

-  The input string is split using a space delimiter.
-  Each resulting token represents one word when empty tokens are excluded.

Vector Size

-  The number of tokens is obtained using `vTokens.size()`.
-  This value becomes the word count.

Reusable Abstraction

-  The function does not need to implement tokenization again.
-  It reuses the previously created `SplitString()` utility.

Procedures

1. Read the complete sentence.
2. Call `SplitString()` using `" "` as delimiter.
3. Store the returned tokens.
4. Obtain the vector size.
5. Display the word count.

💡 Example

Input:

Sara Alterkawi

➡ Result

The number of words in your string is: 2

Note

- ◆ Correct handling of multiple spaces depends on whether `SplitString()` ignores empty tokens.
- ◆ Leading and trailing spaces should be handled consistently.
- ◆ An empty string should normally produce a word count of `0`.

⊘ Warning

- ◆ Counting vector elements is correct only when the tokenization function produces valid word tokens.
- ◆ Multiple consecutive delimiters can create empty strings.
- ◆ Punctuation remains attached to tokens unless it is explicitly processed.



Comparison

▨ `CountWords()` via `SplitString()`

- ✓ Reuses an existing tokenization function.
- ✓ Produces a clean and modular design.
- ✓ Requires vector storage for the tokens.

▤ Character-State Word Count

- ✓ Scans the string directly.
- ✓ Can use $O(1)$ additional space.
- ✓ Requires careful handling of transitions between spaces and words.

✦✦ Summary

- ✓ Converts a sentence into word tokens.
- ✓ Uses vector size as the word count.
- ✓ Demonstrates function reuse and abstraction.
- ✓ Shows how tokenization can support higher-level string operations.

Thank you 😊